

Neoplasia

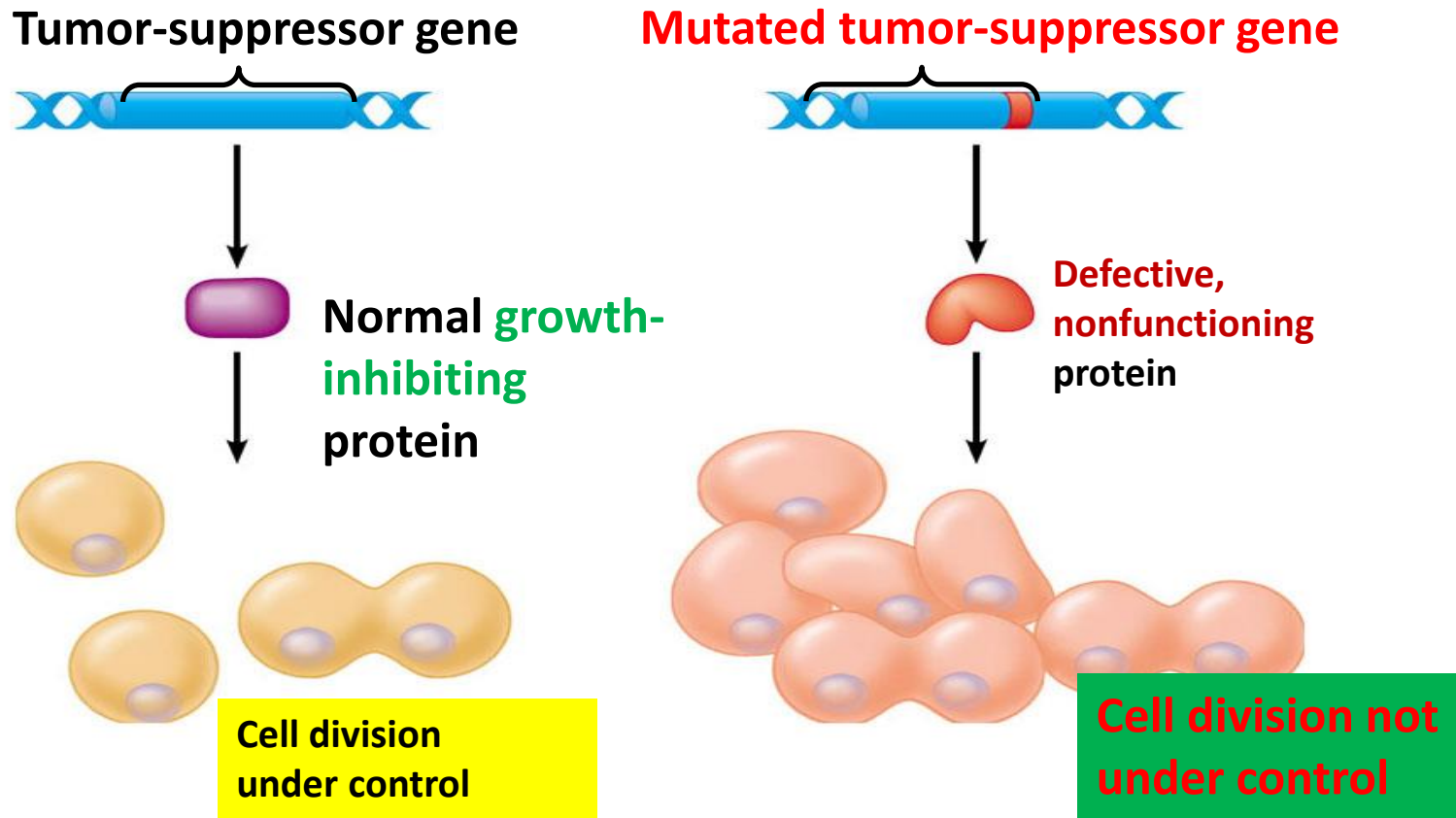
**Microorganisms as agents of
human cancer**

Genes and Cancer

- Proto-oncogenes
 - Genes that **create proteins** that normally activates cell division:
 - growth factor genes
 - become oncogenes (cancer-causing) when mutated
- Tumor-suppressor genes
 - normally inhibit (turn “off”) cell division
 - if switched “OFF” (e.g. due to mutations) ➔ can **cause cancer**
 - example: p53 gene

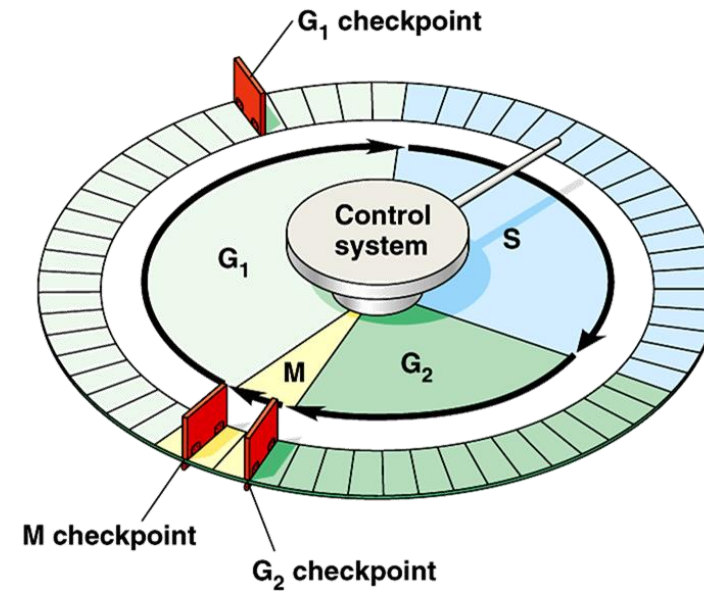
- ***Tumor-Suppressor Genes***

- Mutations that inactivate tumor suppressor genes:
 - Have similar effects as oncogenes



Cancer & Cell Growth

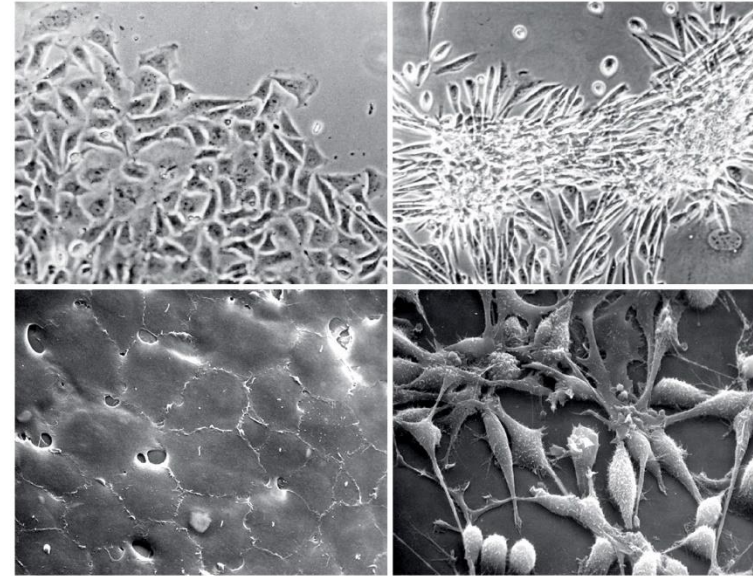
- Cancer = **failure of cell division control**
- What control is lost?
 - lose checkpoint “**stops**”
- Gene **p53** plays a key role in G₁ checkpoint
- p53 protein STOPS cell division if it detects damaged DNA
 - Options:
 - stimulates repair enzymes to fix DNA → forces **cell into G₀ resting stage**
 - keeps cell in G₁ arrest → **causes apoptosis of damaged cell**
 - **50%** cancers have a mutation in p53 gene



- M- mitosis
- G₁ - cells grow
- S - DNA synthesis
- G₂ - growth and preparation for mitosis
- **G₁/S decision point for going to dividing state**

Transformation and potential tumorigenesis

- **Transformation**
 - **alteration** in a **cell's properties** that leads to:
 - **immortalization** and
 - **different growth patterns** that result from alteration in cell cycle
 - Loss of anchorage dependence
 - Loss of contact inhibition (foci)
 - Decreased requirements for growth factors



Development of Cancer

- Cancer develops when mutations accumulate:
 - unlimited growth
 - turn on growth promoter genes
 - ignore checkpoints
 - turn off tumor suppressor genes (p53)
 - escape apoptosis
 - turn off suicide genes
 - immortality = unlimited divisions
 - turn on chromosome maintenance genes
 - promotes blood vessel growth (angiogenesis)
 - turn on blood vessel growth genes
 - overcome anchor & density dependence
 - turn off touch-sensor gene

Association of infectious agents with cancer

Three different importance of infectious agents to human cancer :

1. As direct causes of cancer
2. As source of co-infection
3. Their potential as therapeutic agents against cancer

- When eukaryotes and multicellular organisms appeared over a billion years ago →
- co-evolution with microbes [**eubacteria**, archaea, protists, fungi, viruses] → → establishing persistent relationships:
 - mutualism,
 - commensalism,
 - parasitism
- clearly written in the **genomes of eukaryotic hosts** and their **associated microbiota**:
 - The unique microbial genes in our body → **outnumber human genes** (X₁₀₀)
 - However, many microbial genes are **functionally redundant**
 - we are symbionts or **metaorganisms** **composed of host and microbial cells**

- cross talk between microbes and host cells [via **small molecule metabolites**, **nucleic acids**, and **physical protein-protein interactions**] is necessary for:
 - health,
 - survival, and
 - regulation of physiological functions

locally,
at the barrier level, and
systemically.
→
microbe-dependent
host physiology and
tissue homeostasis

including
• **innate**
• **adaptive resistance** to infections
• **tissue repair**.

Mostly because of its effects on
metabolism,
cellular proliferation,
inflammation,
and immunity,

the microbiota regulates cancer at the level of:

- Cancer predisposing conditions,
- Cancer initiation,
- genetic instability,
- susceptibility to host immune response,
- progression,
- comorbidity, and
- response to therapy

Mechanisms of Infection-Induced Malignancy

1. Chronic Inflammation and Carcinogenesis

- **Chronic host-pathogen interaction**
 - Immunosuppression
- **Chronic inflammation**
 - Oxidative Stress
 - Chemokines/cytokines
 - DNA damage and mutations
 - Cell injury

2. Production of Oncogenic Proteins

3. Genomic Instability from Viral Genomic Integration

Basic facts about inflammation and cancer

1. Chronic inflammation increases **cancer risk**.
2. **Subclinical, often undetectable, inflammation** may be **as important** in increasing cancer risk (for instance, obesity-induced inflammation).
3. Various types of **immune** and **inflammatory cells** are **frequently present within tumors**.
4. **Immune cells affect malignant cells** via production of cytokines, chemokines, growth factors, prostaglandins and reactive oxygen and nitrogen species.
5. **Inflammation impacts** every single **step of tumorigenesis** (from initiation through tumor promotion, all the way to metastatic progression).
6. In developing tumors **anti-tumorigenic** and **pro-tumorigenic immune and inflammatory** mechanisms **coexist**, but if the tumor is not rejected, the pro-tumorigenic effect dominates.
7. Signaling pathways that mediate the pro-tumorigenic effects of inflammation are often subject to a feed-forward loop (for example, activation of NF- κ B in immune cells induces production of cytokines that activate NF- κ B in cancer cells to induce chemokines that attract more inflammatory cells into the tumor).
8. Certain immune and inflammatory components may be dispensable during one stage of tumorigenesis but absolutely critical in another stage

HUMAN CARCINOGENIC MICROORGANISMS

- **Infectious agents** play an etiologic role in approximately **20% of cancer** cases worldwide
- ➔ **[Eleven pathogens]**
 - directly via the expression of their protein products **OR**
 - indirectly via chronic inflammation.
- **Seven viruses:**
 - Hepatitis virus B (HBV)
 - **Hepatitis virus C (HCV)**
 - human papillomavirus (HPV)
 - Epstein–Barr virus (EBV)
 - **human T-cell lymphoma virus 1 (HTLV-1)**
 - Merkel cell polyomavirus (MCPyV) and
 - Kaposi's sarcoma virus also known as human herpes virus 8 (KSVH or HHV8)

- HPV,
- HTLV-1,
- EBV,
- MCPyV, and
- KSVH

direct carcinogenic pathogens and share several similarities

- A **critical portion of the viral genome** can generally be **detected in each cancer** cell ➔ ➔
- **resulting** in the **expression of viral oncogenes** that:
 - disrupt cell-cycle checkpoints,
 - inhibit apoptosis, and
 - contribute to cell immortalization

Indirect sources of cancer

- **One bacterium**
 - **Helicobacter pylori**
- **Three parasites**
 - Schistosoma haematobium,
 - Opithorchis viverrini, and
 - Clonorchis Sinensis

- The indirect carcinogenic pathogens:

- HBV,
- HCV,
- H. pylori,
- S. haematobium,
- O. viverrini, and
- C. sinensis,

do not **induce**
expression of
oncogenes

- Their **persistent infection** leads to a **chronic inflammatory state**.
- Persistent inflammation → the release of:
 - chemokines,
 - cytokines,
 - prostaglandins, and
 - reactive oxygen species,

- Result in
 - the **deregulation** of the **immune system** and
 - **promotion** of **neovascularization**

- *H. pylori* [**non-cardiac gastric carcinoma**]
- HBV [associated with hepatocellular carcinoma (**HCC**)]
- HCV [associated with hepatocellular carcinoma (**HCC**)]
- High risk oncogenic **HPVs**
 - **ano-genital cancers** including **Cervical Cancer**
 - a **subset of head and neck cancers**, and
 - **skin cancers**
- HBV and HCV establish **chronic liver infection** and account for **over 80% of HCCs**
- In some patients, the **immune response to chronic viral infection progresses** to **fibrosis and cirrhosis** → **ultimately HCC**.
- The herpesvirus Epstein-Barr virus (**EBV**):
 - is the causative agent of **Burkitt lymphoma**,
 - **nasopharyngeal carcinoma**, and
 - a subset of **gastric carcinomas**

for over 90% of all infection-attributed cancers

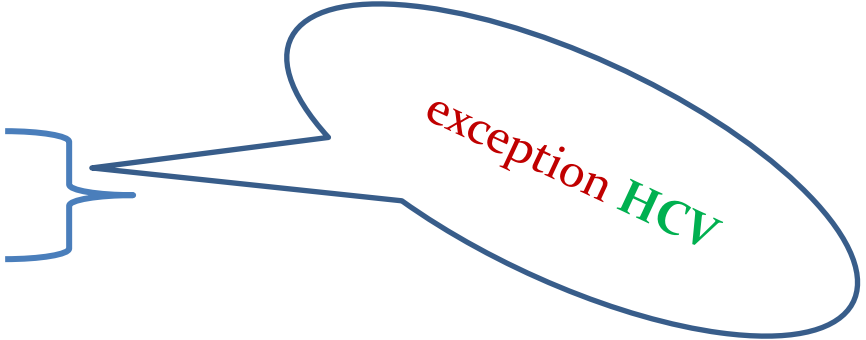
The **Kaposi sarcoma-associated herpesvirus (HSV8)**

- causes Kaposi sarcoma and other pathologies in **immunosuppressed** or elderly individuals

- Human T-cell lymphotropic virus type 1 (HTLV-1)
 - associated with the **T cell lymphomas** prevalent in **certain geographical regions**
- Merkel cell polyomavirus (MCPyV)
 - associated with the majority of cases of **Merkel cell carcinoma**,
 - an aggressive **skin cancer** observed in **immunosuppressed individuals**

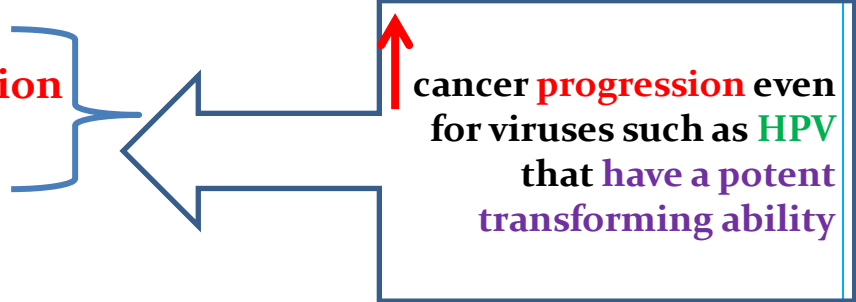
- **All human oncogenic viruses**

- **encode at least one oncogene** and
- **directly induce cell transformation**



exception **HCV**

- Factors such as
 - infection-associated **genital tract inflammation**
 - **vaginal dysbiosis**



cancer **progression** even
for viruses such as **HPV**
that **have a potent**
transforming ability

The Role of Viruses in Tumorigenesis

How do viruses exert their carcinogenic effects?

Phases of tumorigenesis

- Initiation
- Promotion
- Progression

- **On the early stage of oncogenesis:**

- Involving early stage of genetic events linked to
 - Tumor initiation
 - viral integration and insertional mutagenesis
 - viral promotion of DNA damage
 - dysregulation of cellular processes by viral proteins

- **At later stages by modulating:**

- signaling pathways involved in regulation of cell proliferation
- Apoptosis
- Replicative immortality
- Tumor promotion

Alteration of four broad categories of cancer-related genes

- Defective DNA repair genes
- Activation of oncogenes
- Inactivation of tumor suppressors (that affect cell proliferation)
- Reduction/inhibition of genes involved in apoptosis



Slowly builds up numerous mutations (up to 80) in cancer-related genes

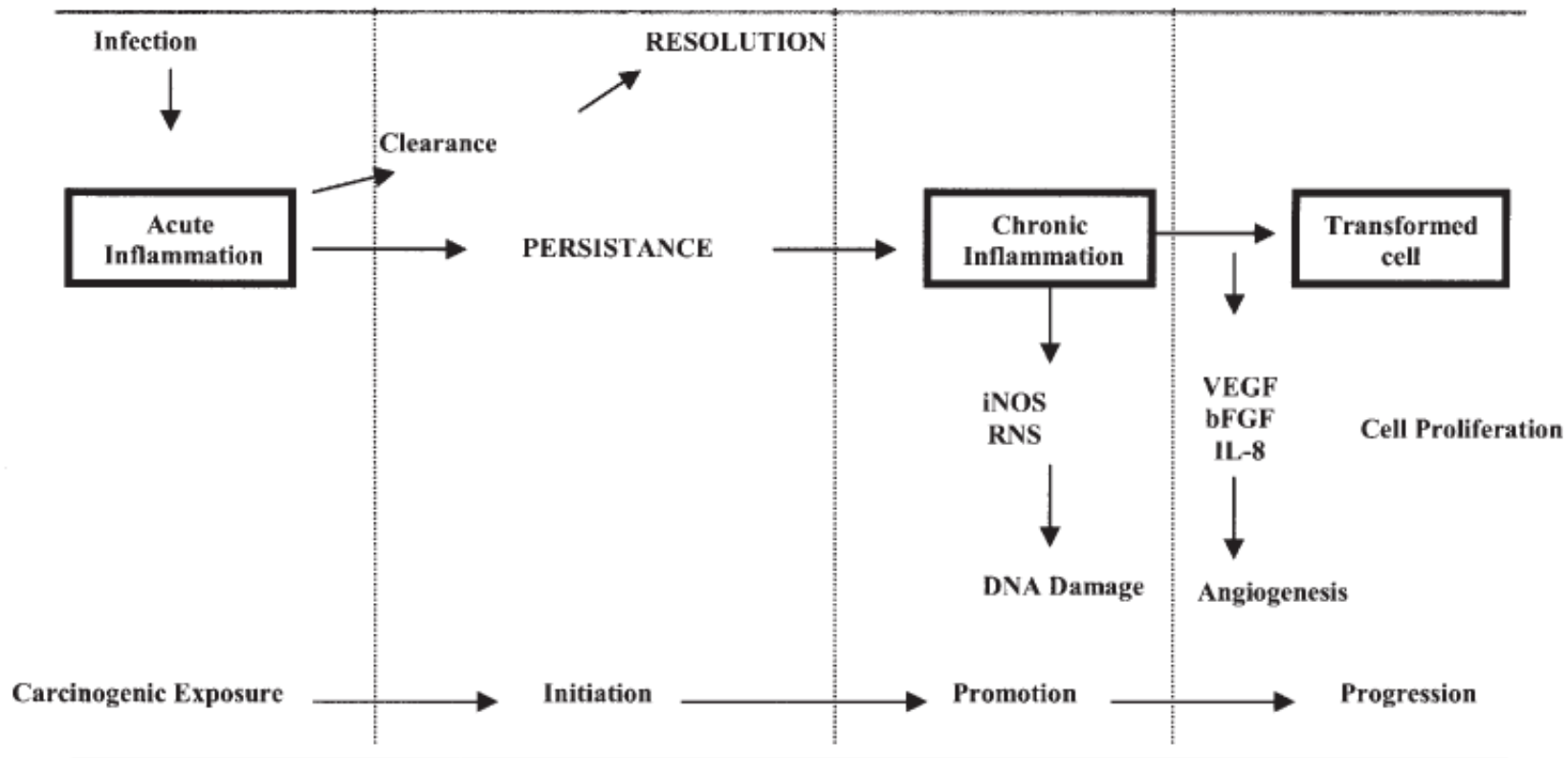


Figure 1. Sequence of events of infection/inflammation and cancer. Acute inflammation occurs, as the host's first response to eliminate an infection. Unresolved infection may result in chronic inflammation. Inflammation generates the production of iNOS and RNS, which are potent DNA-damaging agents causing genome mutations and instability. Chronic inflammation contributes to apoptosis and tissue generation with the promotion of microvessel formation and cell proliferation. This sequence of events corresponds to the stages of initiation, promotion and progression for cancer development.

Human Oncovirus Replication and Persistence Strategies

Find/ Create conditions for replication

- ◆ Induce the Cell cycle
- ◆ Metabolic reprogramming
- ◆ Inducing angiogenesis

Ensure correct replication

- ◆ Recruit or inhibit DDR

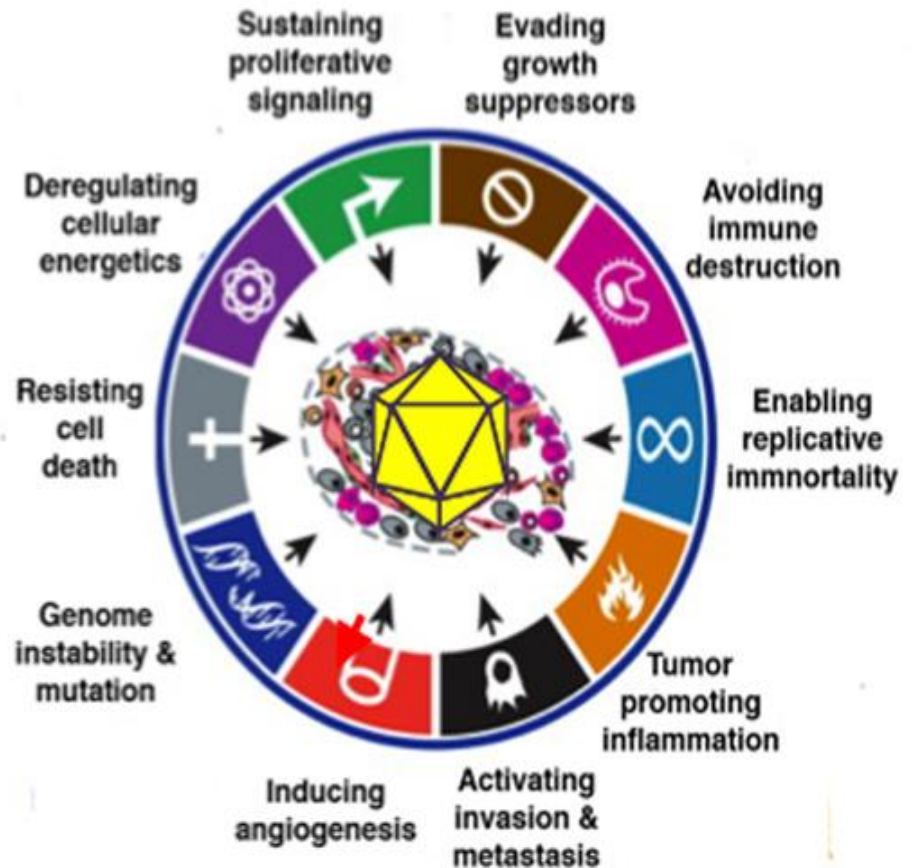
Maximize virus production

- ◆ prevent apoptosis until virion matures
- ◆ Immune evasion

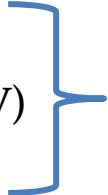
Multiply latent episomes or provirus

- ◆ Cell survival
- ◆ Cell immortalization
- ◆ Cell proliferation

Hallmarks of Cancer



Important points about Viral oncogenesis

1. Viruses are responsible for over 12% of the world's cancers.
2. Up to 80% of these human viral-associated cancers are:
 - cervical cancer (associated with human papilloma virus or HPV) and
 - liver cancer (associated with the hepatitis B virus or HBV and the hepatitis C virus or HCV).
3. Besides being oncogenic on their own right,
 1. Epstein-Barr virus (EBV)
 2. the Kaposi sarcoma-associated herpesvirus (KSHV)
 3. human T-lymphotropic virus type I (HTLV-I)

**increase the risk of
certain cancers.**
4. Human immunodeficiency virus (HIV), ➔ **increase the risk of certain cancers.**
5. The **development of tumors** is a **multistep** process depending on the **accumulation of mutations altering a number of genes**.
6. **Most virus-associated cancers have long latency periods of several decades**
7. Only a small percentage of the people infected with the virus actually develop the cancer.

Summary: Human viruses associated with cancer development

Specific Viruses	Cancers Caused	Mechanism that causes cancer
Epstein-Barr Virus (EBV)	Burkitt's Lymphoma, Hodgkin's Lymphoma	Cell proliferation, inhibition of apoptosis, cell migration
Hepatitis B Virus (HBV)	Hepatocellular Carcinoma	Inflammation; insertional mutagenesis
Hepatitis C Virus (HCV)	Hepatocellular Carcinoma	Inflammation
Kaposi's Sarcoma virus	Kaposi's Sarcoma, Lymphoma	excessive level of vascular endothelial growth factor
Human Papilloma Virus (HPV)	Cervical Cancer	Inhibition of DNA damage response, anti-apoptotic activity
Merkel cell polyomavirus (MCPyV)	Markel cell carcinoma (MCC)	Retinoblastoma (RB) inactivation; viral oncogenic products; cellular oncogene mutation due to viral DNA integration; damage DNA Damage Response (DDR)
Human T-cell Lymphotropic Virus	Adult T-cell Leukemia and Lymphoma	Immortalization and transformation of T-cells

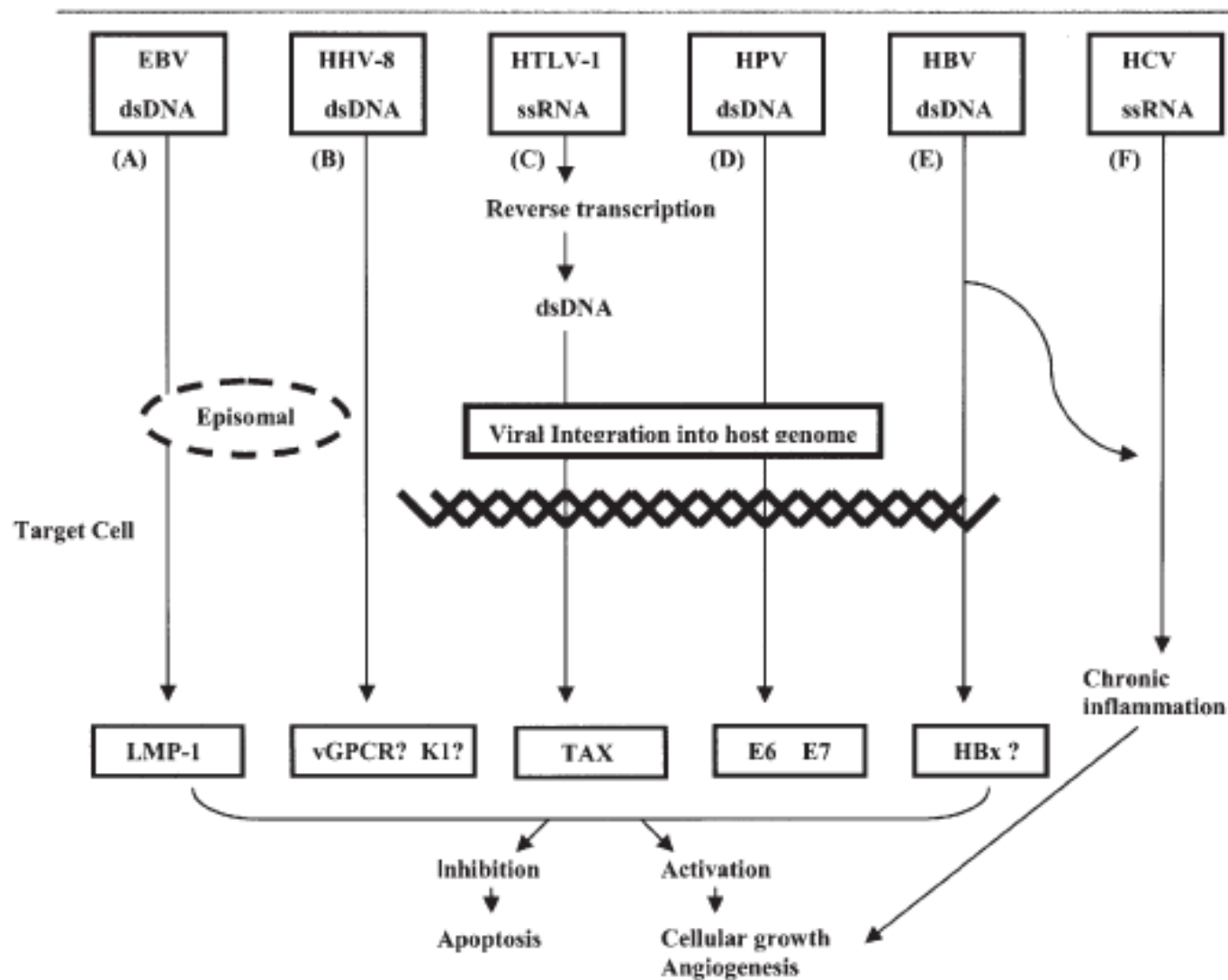


TABLE 1 Human cancer viruses: transformation and tumorigenicity

Virus	Abbreviation(s)	Type	Family	Cancer(s)	Tropism(s)	Integration	Infectious virus produced	Oncoproteins
Human papillomaviruses 16 and 18	HPV16, HPV18	DNA	Papillomavirus	Cervical carcinoma, anal carcinoma, penile carcinoma	Keratinocytes	Episomal and integrated	Yes	E6, E7
Epstein-Barr virus	EBV, HHV-4	DNA	Herpesvirus	Burkitt's lymphoma, nasopharyngeal carcinoma	B lymphocytes, epithelial cells	Episomal	Yes	EBNA-1, EBNA-2 LMP-1, LMP-2 EBERs, BARTs
Kaposi's sarcoma-associated herpesvirus	KSHV, HHV-8	DNA	Herpesvirus	Kaposi's sarcoma, primary effusion lymphoma, multicentric Castleman's disease	B cells, macrophages, keratinocytes, endothelial cells, etc.	Episomal	Yes	LANA, vIL-6, vMIP-1, vFLIP, vBCL-2, v-cyclin-D, vGPCR, vIRF-1
Hepatitis B virus	HBV	DNA	Hepadnavirus	Hepatocellular carcinoma	Liver	Yes	Yes	HBx, HBsAg
Hepatitis C virus	HCV	RNA	Flavivirus	Hepatocellular carcinoma	Liver	No	Yes	Core protein, NS3
Human adult T-cell leukemia virus type 1	HTLV-1	RNA	Retrovirus	T-cell leukemia	T lymphocytes	Yes	Yes	Tax, p12, p30 p8, p13
Merkel cell polyomavirus	MCV, MCPyV	DNA	Polyomavirus	Merkel cell carcinoma	Merkel cells	Yes	No	T antigen, t antigen

Immune evasion of virally-induced carcinogenesis?

1. Down regulation of
 - Major Histocompatibility complex or associated Functions (EBV, HBV)
 - Interfering with antigen processing and presentation (HPV's E5)
2. Interfering with interferon action
3. Molecular mimicry
4. Generation of escape mutants
5. Expression of viral proteins at very low level
6. Infecting cells away from immune surveillance

Hepatocellular carcinoma (HCC) due to HBV and HCV

- HBV and HCV establish chronic liver infection and account for over 80% of HCCs
- In some patients, the immune response to chronic viral infection progresses to:
 - fibrosis and
 - cirrhosis and ultimately → → HCC.
- Initial innate response to **HBV** infection is modest, **BUT**
- **HCV** actively evades innate responses by inhibiting
 - production and the signaling of interferon
- HBV may **directly transform** hepatocytes.
- HCV transforms hepatocytes **indirectly**
- In both HBV and HCV viruses,
 - the **pathogenesis** of **HCC** is largely dependent on immune-related inflammation

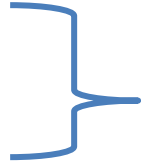
HBV

- The **hepatitis B virus (HBV)** is an enveloped DNA virus that may potentially cause chronic hepatitis in those infected.
- There is a strong link between chronic infection with HBV and hepatocellular carcinoma,
 - which typically appears after **30-50 years of chronic liver damage** and liver cell replacement.
- Chronic carriers of HBV have a 300 times greater risk of eventually developing liver cancer.
- Around
 - 90% of individuals infected at birth and
 - 10% of individuals infected as adults become chronic carriers of HBV.
- Worldwide, HBV is responsible for 60% of all liver cancer cases.

HBV-associated carcinoma

- HBV-associated carcinogenesis can be seen as a multi-factorial process that includes

- **both direct** and
- indirect mechanisms



might act
synergistically.

1. The **integration of HBV DNA** into the host genome:

- occurs at early steps of clonal tumor expansion
- exert insertional mutagenesis;
- gene and chromosomal deletions and translocations,
- amplification of cellular DNA, and

2. **The permanent liver inflammation**

- results in a degeneration and regeneration process
- confers to the accumulation of critical mutations in the host genome

3. The **regulatory proteins HBx** and the PreS2 activators

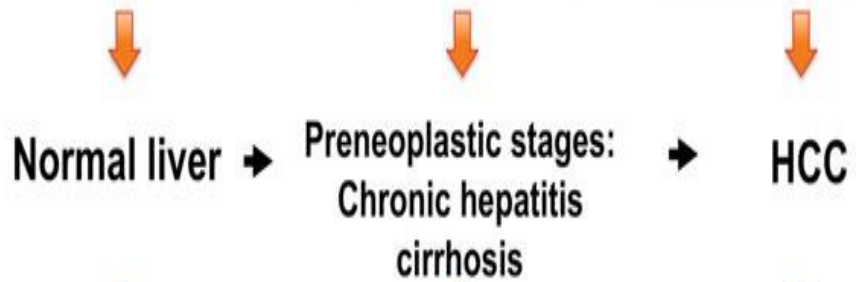
- that can be encoded by the integrated genome
- exert a tumor promoter-like function,
- resulting in positive selection of cells producing a functional regulatory protein

- **multiple pathways are involved in the process of carcinogenesis, including:**
 - the **accumulation of genetic damage** due to **immune-mediated hepatic inflammation**
 - the induction of **oxidative stress**
 - a virus-specific mechanisms involving the viral proteins HBx and HBs
 - the **insertional mutagenesis** with **integration of HBV DNA** into the host genome: ➔
 - that **alters the expression of endogenous genes** in genes regulating
 - cellular **signal transduction cascades**,
 - proliferation control and
 - cell viability
 - induces **chromosomal instability**
 - induce deletions in the host chromosome at the integration site
 - **epigenetic modification** through the modification of the **genomic methylation** status and also the regulation of microRNA (miRNA) expression



HBV DNA integration
HBx and envelope
glycoproteins
Chromosomal
instability
Epigenetic lesions

Clonal genetic lesions
Mutations
Chromosome alterations



Host factors:
Gender, age
Cooperating risk factors

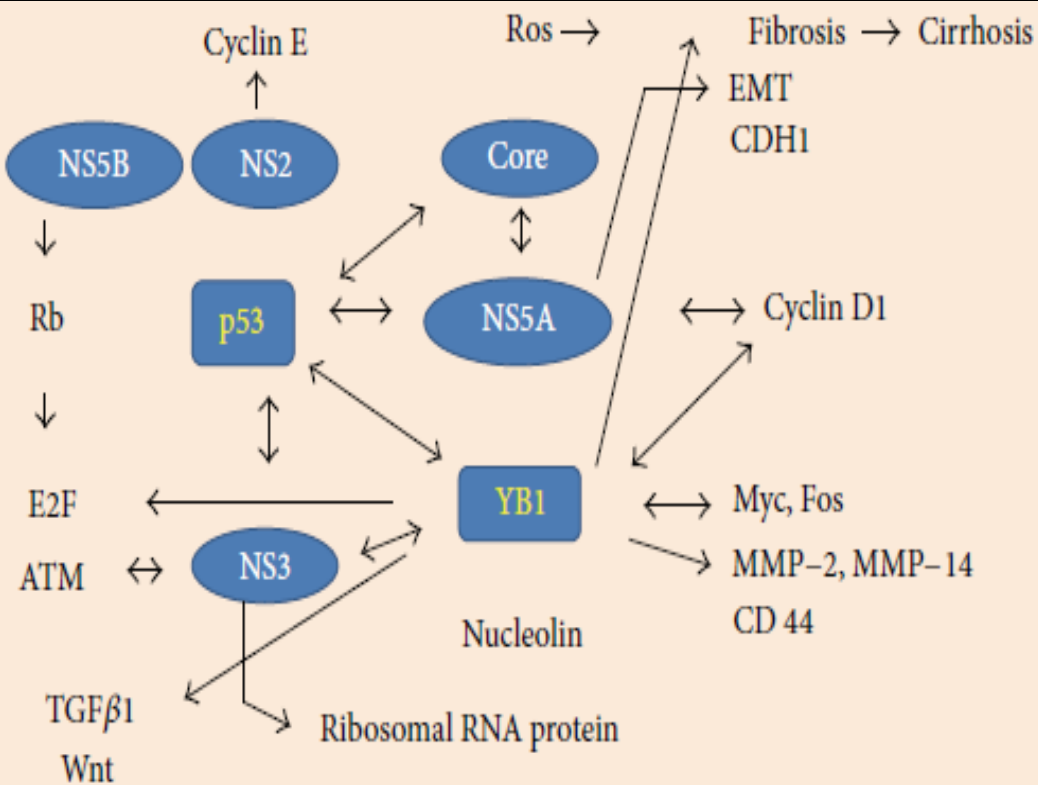
Inflammation
Oxydative stress
Apoptosis/necrosis
Regeneration

Familial predisposition
Polymorphisms

- **Multistep process** of HBV-induced liver tumorigenesis:
 - **Chronic HBV infection**, **alone** or **in combination with cofactors** → triggers a cascade of events ultimately **leading to transformation**.
- At pre-neoplastic stages,
 - the immune response induces a **neco-inflammatory disease** that creates a **suitable environment** for the occurrence of **genetic and epigenetic lesions**.

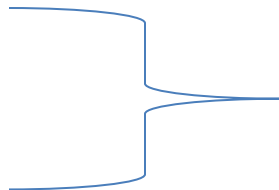
- **Integration of HBV DNA sequences** into the host genome as well as **long-term production of the viral HBx** and **surface proteins** may contribute to the **tumoral process via multiple mechanisms**

- The **hepatitis C virus (HCV)** is
 - an **enveloped**
 - **RNA virus**
 - **chronic hepatitis** in those infected.
- Around **85% of individuals infected** with HCV become **chronic carriers** and
- As with HBV, there is a **strong link** between **chronic infection** with HCV and **liver cancer**,
 - typically **appearing after 30-50 years of**
 - **chronic liver damage** and
 - **liver cell replacement.**
- HCV-initiated HCC development may be driven at least in part by **chronic immune-mediated inflammation.**
- Worldwide, HCV is responsible for **22 % of all liver cancer cases.**



- **Multiple interactions between HCV-encoded proteins and**
 - **host cell cycle regulators and**
 - **tumor suppressor proteins**
- three distinct HCV proteins,:
 - core ,
 - NS₃, and
 - NS_{5A}, interact with the p53 tumor suppressor

- Viral proteins
 - Core,
 - NS₂,
 - NS₃, and
 - NS_{5A}

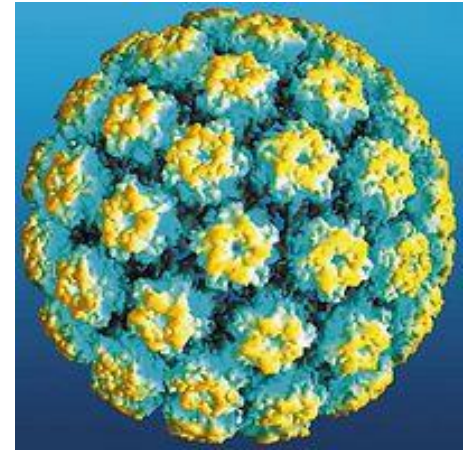


- **subvert the normal functioning of many crucial cellular proteins**
 - such as p53 and YB-1 and
 - oncogenic products such as Myc and Ras, → **transforming a normal cell into a cancer cell**

- NS_{5A} **interacts with many proteins** implicated in
 - signal transduction,
 - cell growth and
 - death,

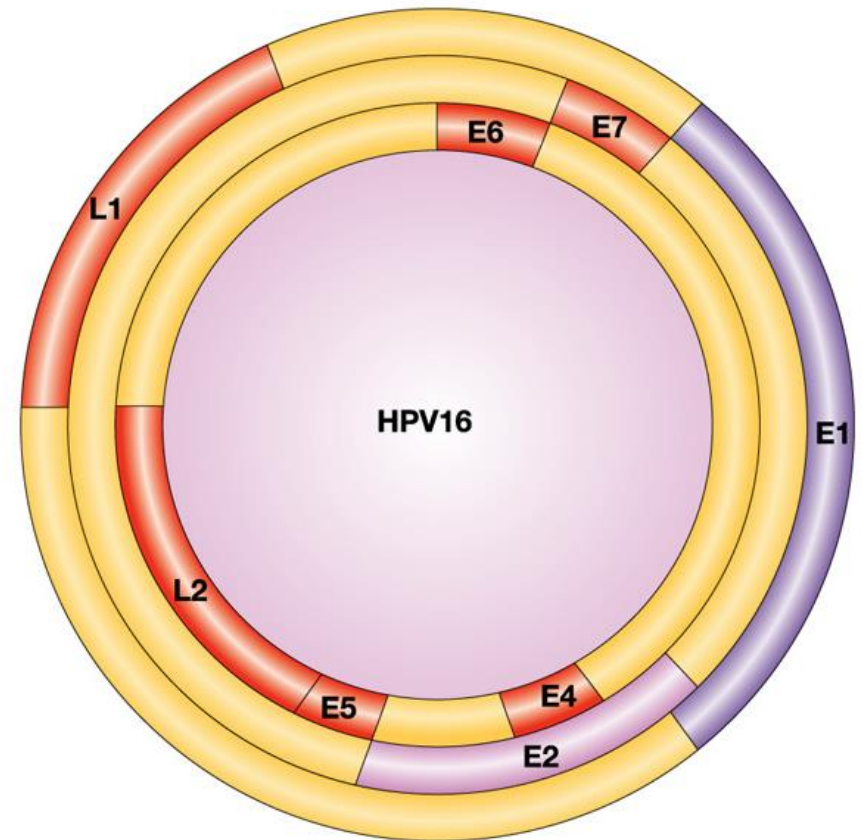
The Human Papilloma Viruses (HPV)

- They are members of the **Papillomaviridae** family,
 - a large family with a **tropism for squamous epithelium**
- Human Papillomaviruses (HPVs) :
 - small,
 - Naked
 - Icosahedral
 - double-stranded DNA viruses.
- The **capsid** is composed of:
 - mainly one type of protein called **major capsid protein, L₁**, and
 - small amount of **minor protein, L₂**
- **Eight genes** in
 - a circular,
 - double-stranded DNA genome, ➔ characterized as **either early (E)** or **late (L) genes**:
 - early genes are
 - produced early in the virus life cycle.
 - associated with DNA replication, regulatory functions, and activation of the lytic life cycle,
 - the late genes are involved with production of the viral capsid parts.
 - LCR (long control region)



Human papillomavirus

- E1 and E2: minimal gene expression, suppress expression of E6 and E7
- E6: prevents cell differentiation and promotes p53 degradation
- E7: prevents cell-growth arrest/differentiation
- L1, L2: capsid proteins



The human papilloma viruses (HPV)contd.

- The different HPV types have been categorized into five genera:
 - Alpha
 - Beta
 - Gamma
 - Mu
 - Nu
- More than **200 different HPV types** have been identified
 - Based on the tissue they infect:
 - **cutaneous** and
 - **mucosal** HPVs
- Alpha PVs responsible for all cases of
 - **cervical cancer**
 - cancer of the **head and neck**,
 - **vulva**,
 - **vagina**,
 - **penis**, and
 - **anus**.
- Also responsible for warts.
 - Warts are generally considered as benign tumors,
 - several types of warts, including in the genital region

9/21/2022 In these tumor cells the viral DNA is usually found integrated in host cell chromosomes.

Main HPV genotypes and their associated diseases

Genus	Species	Representative HPV types	Tropism	Associated Diseases
Alpha-PV	$\alpha 1$	32,	Mucosal	Heck's disease
	$\alpha 2$	3, 10, 28	Cutaneous	flat warts
	$\alpha 4$	2, 27, 57	Cutaneous	Common warts
	$\alpha 7$	18, 39, 45, 59, 68	Mucosal	Intraepithelial neoplasia, invasive carcinoma
	$\alpha 9$	16, 31, 33, 35, 52, 58	Mucosal	Intraepithelial neoplasia, invasive carcinoma
	$\alpha 10$	6, 11	Mucosal	Condylomata acuminata
		13		Heck's disease
Beta-PV	$\beta 1c$	5, 8, 12, 14, 19, 20, 21, 24, 25, 36, 47	Cutaneous	Epidermodysplasia Verruciformis
	$\beta 2$	9, 15, 17, 22, 23, 37, 38	Cutaneous	Epidermodysplasia verruciformis
	$\beta 3$	49	Cutaneous	Epidermodysplasia verruciformis
Gamma-PV	$\gamma 1$	4, 65	Cutaneous	Warts
	$\gamma 4$	60	Cutaneous	Warts
Mu-PV	$\mu 1$	1	Cutaneous	plantar warts
	$\mu 2$	63	Cutaneous	Warts
Nu-PV	ν	41	Cutaneous	Warts

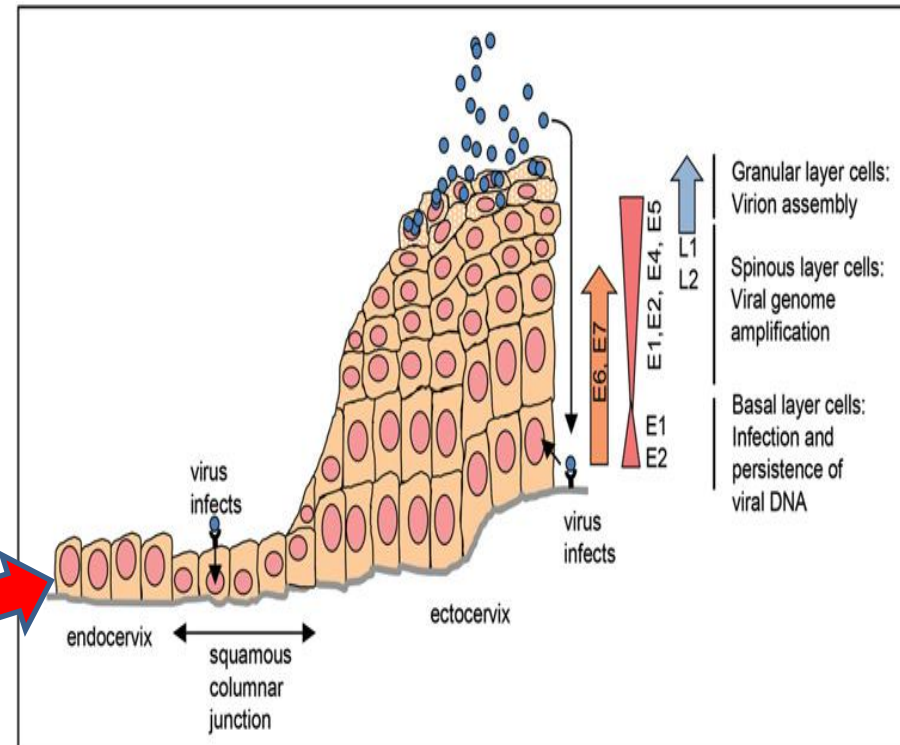
- The mucosal HPVs are clinically classified as:
 - “high-risk”
 - high-risk HPVs cause intraepithelial lesions that are at risk for malignant progression.
 - Infection with high-risk HPVs are associated with approximately 5% of all human cancers, in particular with cervical carcinoma, the third most common cancer in women worldwide
 - Anogenital neoplasia:
 - unusual growth of cells in the genital and/or anal regions caused by high-risk (HR) types,
 - » types 16, 18, 31, 33, 35, 39, 45, 51, 52, 56, 58, 59, 68, 73, and 82
 - “low-risk” based on the ability to effect cellular transformation.
 - Low-risk HPVs, such as HPVs 6 and 11, cause benign genital warts,

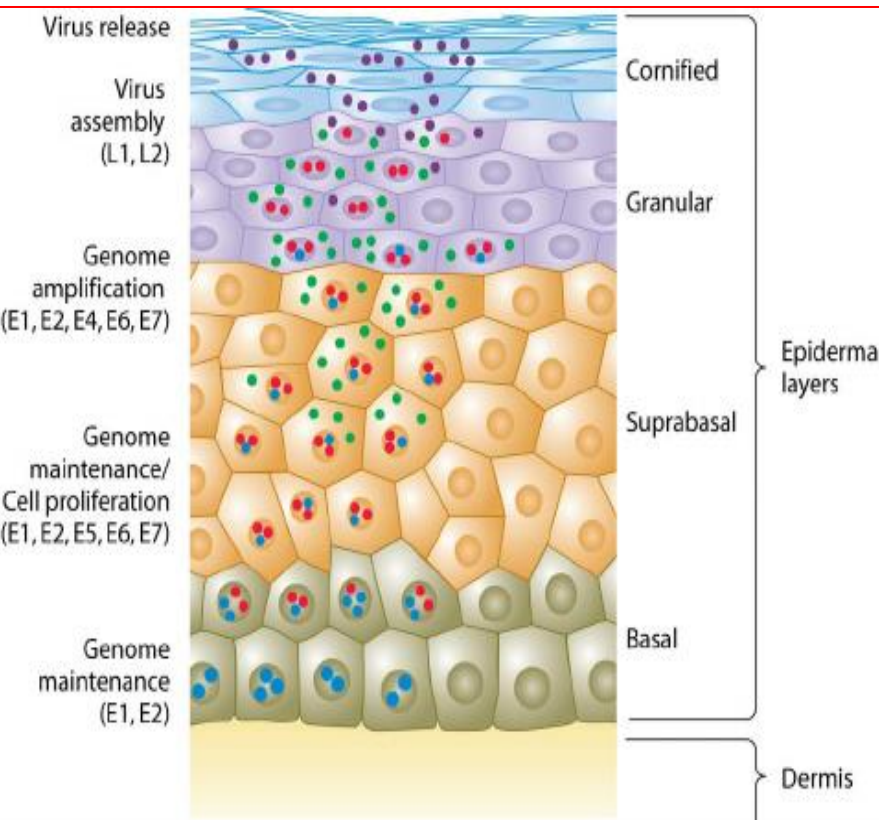
Viral strategies to evade or subvert immune attack

- Keeping very low profile
 - Non-secreting proteins
 - No viraemia
 - No lysis
 - limited antigen production
 - HPV 16 E7 inhibits interferon regulatory factor 3
 - HPV 18 E6 inhibits the JAK-STAT activation response
 - reduced inflammatory response
 - E5 can modulate antigen processing pathways
-  Influence the polarization of Th cell types

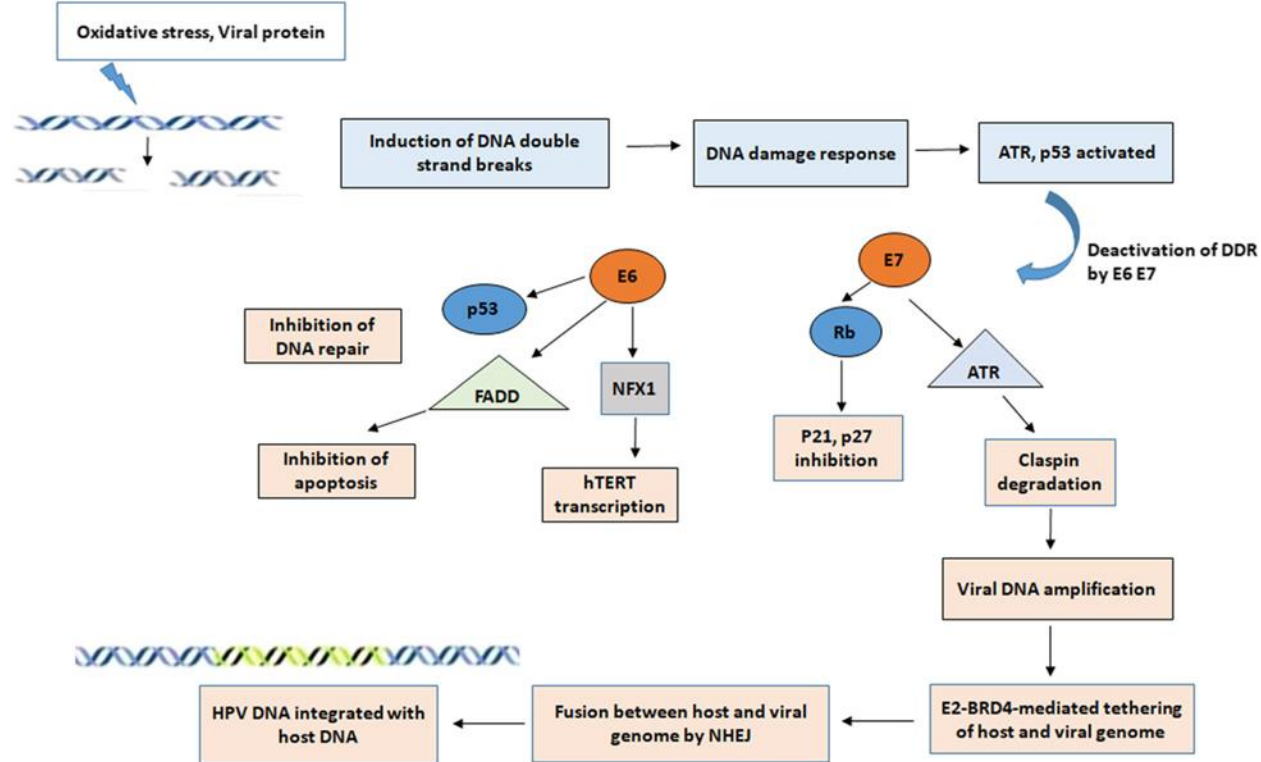
Life Cycle of High-Risk HPVs in Cervical Epithelium.

- The progression of the life cycle of HPV depends on several different factors, such as:
 - differentiation of the host cell,
 - type of HPV, or
 - host immune system.
- HPV can **only infect parabasal** or **basal cells**
 - **undifferentiated stem cells** of the epidermis
 - **most commonly** in the **transformation zone** of the cervix,
 - **adjacent to the border of the endocervix and ectocervix.**
- This area becomes **more accessible** during
 - **puberty**
 - **pregnancy** and
 - **when using oral contraceptives.**
- HPV infection requires the presence of a **micro-wound** that allows the infectious virions to **access the basal lamina.**



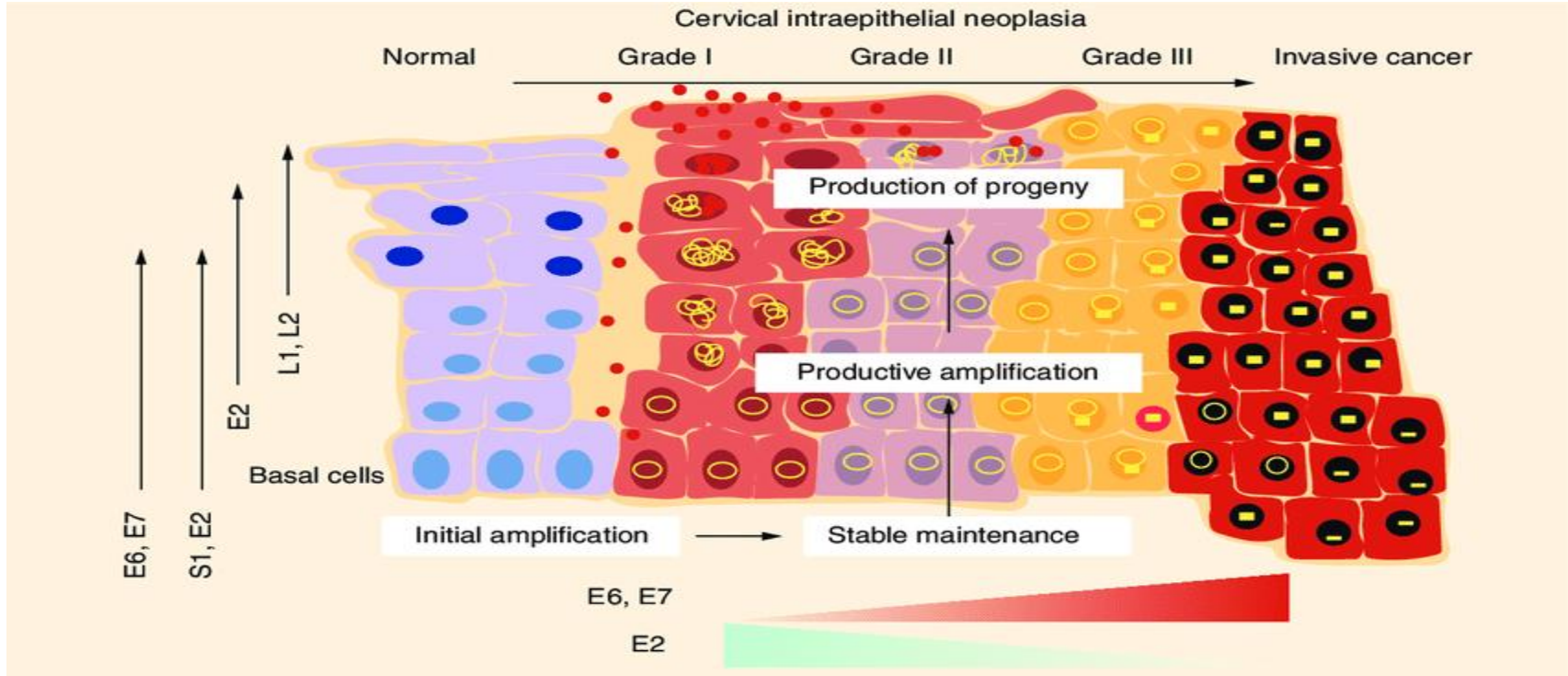


- The infected basal cells form the reservoir of infection, → Initial replication rapid → then to stable number →
 - the viral genome is **maintained** as a low copy number **episome** [from 50 to 200 or so copies per cell] →→ called **maintenance replication**
- There is a differential **spatial** and **temporal pattern of HPV gene expression** in the infected epithelium
- As these cells divide, they produce daughter cells that are pushed outwards towards the epithelial surface.
- In lesions (such as CIN₁) caused by high-risk HPV types (such as HPV₁₆), cells in the lower layers express E6 and E7 and are driven through the cell cycle
- In the mid layers, proteins necessary for genome amplification become elevated in these cells → allowing genome amplification to occur.
- These cells express the viral E4 protein and are typically in the S or G₂ phases of the cell cycle
- In the **upper epithelial layers**, the cells leave the cell cycle, and in a subset of the E4-positive cells, the virus L2 and L1 proteins are made → allowing **packaging of the amplified viral genomes**.



Mechanism of HPV DNA integration:

- Induction of double strand breaks →
 - by reactive oxygen species, →
 - viral protein induces DNA damage response → following which ATR and P53 get activated to repair the damage
- HPV oncogenes deactivate the normal function of DNA damage response (DDR).
 - E7 inactivates Rb → disrupting the cell cycle inhibitors
 - E6 degrades p53 →
 - inhibiting DNA repair,
 - degrades FAS-associated death domain protein (FADD) preventing apoptosis via Fas pathway,
 - promotes degradation of transcriptional repressor, NFX1 and activates hTERT transcription.



The HPV life cycle in association with cancer progression.

- While the copy numbers of the viral genome are maintained constant in the basal cells (stable maintenance),
 - it is drastically increased in a differentiating compartment of the epithelium (productive amplification).
- The dysregulation of the viral gene expression →
 - increase of E6 and E7 levels and decrease of E2 level, →
 - a key event of neoplastic progression of the epithelial lesions with high-risk HPV infection.
- The productive life cycle is supported in low-grade cervical intraepithelial lesions (CIN₁) but aborted in high-grade cervical intraepithelial lesions (CIN₂₋₃)

Other viral-caused tumors

- The **Epstein-Barr virus (EBV)**:
 - Normally causes **benign proliferations** such as **infectious mononucleosis** and **hairy leukoplakia** of the **tongue**.
 - can contribute to serious tumors
 - non-Hodgkin's lymphoma in AIDS patients and
 - post-transplantation lymphoproliferative diseases,
 - appears to be an essential factor for
 - **posterior nasopharyngeal cancer** in some individuals,
 - can be a co-factor for
 - **Burkitt's lymphoma**, and
 - contributes to **smooth-muscle tumors** in **immunosuppressed** children.
- The **Kaposi sarcoma-associated herpesvirus (KSHV)**:
 - transmitted from person to person
 - primarily through saliva,
 - sexually, primarily among men who have sex with men.
 - via blood and
 - Vertical.
 - Healthy individuals infected with the virus show no signs or symptoms.
 - KSHV has been **linked to several cancers**,
 - Kaposi sarcoma and other rare lymphomas.

The **human immunodeficiency virus (HIV)**:

- An immunosuppressive virus responsible for AIDS,
- spread through
 - unprotected sexual activity,
 - infected drug needles,
 - during pregnancy from mother to child, and
 - through infected breast milk.
- a weakened immune system due to HIV increases a person's risk of getting several cancers caused by other viruses:
 - non-Hodgkin lymphomas
 - Hodgkin lymphomas,
 - anogenital cancers,
 - Kaposi sarcoma,
 - oral cancers, and
 - liver cancer.
 - increases the risk of other types of cancers,
 - including eye cancer,
 - non-melanoma skin cancer, and
 - possibly lung cancer.

- The retrovirus **human T-lymphotropic virus type I (HTLV-I)** can induce a rare **adult T-lymphocyte leukemia-lymphoma.**